Program 1

Problem Statement
Overload the '+' and '==' operators for a Point class and overload a 'move' member for a Snake class; run a short loop that uses the overloaded operators to score moves and decrement a life counter until game over.

Source Code:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int score = 0, life = 3;

class Point {
int x, y;
public:
Point() {
x = 0;
y = 0;
}

Point(int a, int b) {
x = a;
y = b;
}

// Overloading + operator
Point operator+(Point p) {
Point t;
t.x = x + p.x;
t.y = y + p.y;
return t;
}

// Overloading == operator to return a direction/relationship code
int operator==(Point p) {
if (x == p.x && y == p.y)
return 1;
else if (p.x > x && p.y > y)
return 2;
else if (p.x > x && p.y < y)
return 3;
else if (p.x < x && p.y < y)
return 4;
else if (p.x < x && p.y > y)
return 5;

return 0;
}
```

```
void show() {
cout << "(" << x << ", " << y << ")" << endl;
}
};

class Snake {
Point P, Q, R;
public:
void move(Point p) {
P = p;
}

void move(Point p, Point q) {
P = p;
Q = q;
}

void play() {
P = Point(rand() % 2, rand() % 2);

while (life > 0) {
Q = Point(rand() % 2, rand() % 2);

cout << "\nP = ";
P.show();
cout << "Q = ";
Q.show();

int a = (P == Q); // Calls overloaded == operator

if (a == 1) {
life--;
cout << "Life = " << life << endl;
}
else if (a == 2) {
score++;
cout << "Up-Up" << endl;
R = P + Q;
cout << "R = ";
R.show();
move(Q);
}
else if (a == 3) {
score++;
cout << "Up-Down" << endl;
move(P, Q);
}
else if (a == 4) {
score++;
cout << "Down-Down" << endl;
move(Q);
```

```
}
else if (a == 5) {
score++;
cout << "Down-Up" << endl;
move(P, Q);
}

cout << "Score = " << score << endl;
}
}
};

int main() {
srand(time(0));
Snake s;
s.play();

cout << "\nGame Over!" << endl;
cout << "Total Score = " << score << endl;

return 0;
}
```

Input:
(no input)

Output (Windows Terminal):

